AWASIR 91

2

PEDIATRICS

كبريت & عميري



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بالتوفيـــــــــق

1# Respiratory system

- Respiratory disorders is the number 1 cause of death in childhood.
- Respiratory infection is the number 1 infection in childhood (80%- 90% viral).
- Increased risk of infections by :

• Host:
- immunodeficiency
- poor nutrition
- underlying lung or heart disease
- male gender

• Environmental:
- parenteral smoking specially maternal
- poor socioeconomic
- overcrowding

Note: most serious time for respiratory infection = infancy

◈ Upper respiratory tract infection (LRTI):

- 80% of respiratory infection
- Ear (otitis media)
- Nose (coryza)
- Throat (pharyngitis, tonsillitis)
- Sinuses (sinusitis)

• **They may cause:**
- febrile convulsions
- feeding difficulties
- acute exacerbation of asthma

❖ Common cold (coryza):	❖ Pharyngitis: at school age
- **Number 1 infection in childhood** - **Rhinovirus** - **usually associated with sinus inflammation = rhinosinusitis** - **it is self-limiting** - **you may give paracetamol or Ibuprofen for fever and pain.** - **nasal saline drop before feeding.** • C/of : - **nasal blockage, discharge** - **no or mild fever**	• viral : adenovirus • bacterial tonsillitis : GAS, EBV you cannot distinguish bacterial from viral but: - marked constitutional symptoms - anterior cervical lymphadenopathy - tenderness - white exudate - headache - no runny nose = Bacterial • **Treatment:** - for bacterial • W**hy?** to prevent rheumatic fever (100%

- **no systemic features** NOTE: **secondary bacterial infection are not common**	prevention if given less than 9 days) **NOTE:** Avoid amoxicillin it is causing rash if EBV is the cause • **What to give?** penicillin V for 10 days Or erythromycin for 10 days Or single IM Benzathil penicillin **NOTE:** Diphtheria can cause both pharyngitis and tonsillitis. if suspected give antitoxin (it is serious). **Complication:** Retropharyngeal abscess Peritonsillar abscess Para tonsillar abscess Ledwing angina = give IV penicillin Cervical lymphadenopathy GAS = rheumatic fever, glomerulonephritis Acute otitis media

❖ Acute otitis media	❖ Otitis media with effusion
- **6 to 12 months** • Why? **short, horizontal, poorly functioning ET (Eustachian tube)** • C/Of : - **otalgia** - **fever** - **red bulging tympanic membrane with loss of light reflex** • Treatment: - **analgesia** - **antibiotic, amoxicillin** - **if there is bus dry by wicking 3/ Day** - **avoid putting anything in the ear** - **follow up in 5 days** - **if resistance give cotrimoxazol** Viral or bacterial • Risks: - **acute mastoiditis** - **meningitis**	- 2-7 years - recurrent otitis media • **C/Of:** - asymptomatic but recurrent dull fluid level. - on Tympanometry = flat trace • **Confirmation :** - if more than 4 years = decrease Hearing (conduction) = audiometry - if less than 4 years = decreased hearing on distraction test. **NOTE:** otitis media with effusion 1st cause of conductive deafness in children, speech problem and learning difficulties. • **Treatment :** - self-limiting - if chronic = Grommet's tube. - if still = Grommet's tube + adenoidechtomy.

NOTE: antibiotic shorten duration of infection but has no effect on hearing loss.

Tonsillectomy :

1- Recurrent tonsillitis
2- Quinsy angina

Tonsillectomy + adenoidechtomy :

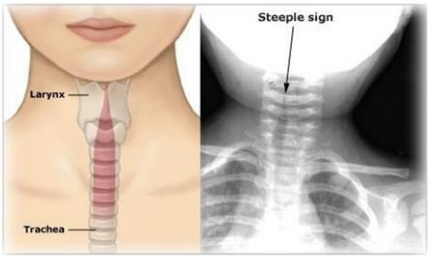
1- Obstructive sleep apnea (absolute indication)
2- Recurrent otitis media with effusion

Laryngeal/ tracheal infections: How to assist severity?

1- Chest indrwing (absent, on crying, at rest).
2- **2-** Stridor (absent, on crying, at rest, biphasic)

How to detect hypoximia? pulse oximetry (most reliable)

❖ Sinusitis	❖ Croup (laryngeao tracheao bronchitis)
- **At birth : ethemoid sinuses, maxillary #1** - **sphenoid later** - **frontal later (no frontal sinuses until more than 10 years).** - **viral : as part of coryza.** - **bacterial : pain, swelling, tenderness.** • Treatment: - **analgesia** - **topical discongestion** - **antibiotic if bacteria** • C/Of : - **persistence of mucopurulent rhinorrhea** - **cough especially at night** - **may exacerbate asthma**	- viral : #1 PIV, RSV - age : 6 months - 2 years - lesions : subglottic stenosis (dangerous) • **C/Of:** - coryza then gradual onset (2-3days) of stridor (harsh voice) - horsness - barking cough (like sea lions) worse at night - No or lower grade fever - no toxicity • **Diagnosis :** - clinically - AP x-ray : subglottic narrowing "Steeple sign" • **Treatment :** - steam inhalation (no proven benefit) - steroids : oral or IM or nebulized - if severe : aerosized epinephrine, temporal relief (2 hours). risk : rebound **NOTE:** striders decrease severity, duration, Hospital stay - relapse : +ve (recurrent croup = like atopy) - intubation : often not needed.

	• #1 **complication :** viral pneumonia  steeple sign (AP View)

❖ Bacterial tracheitis (pseudomembranous croup)

- staph aureus
- croup + fever (high grade) + toxicity + thick respiratory secretions
- **Treatment:**
- IV antibiotic
- intubation may be needed

❖ Acute epiglottitis:

- bacterial : haemophilus influenzae type B. 1- 6 years
- **C/Of :**
- 3rd : toxic, anxious, cherry red epiglottitis.
- acute within hours.
- soft stridor
- no cough
- painful Throat
- drooling of saliva
- inability to swallow
- decrease in speaking
- high grade fever
- toxicity

sniffing position : up right, open mouth, forward leaning.

NOTE: cherry red epiglottitis seen without introduced spatula

- **Treatment :**
- 1st : secure Airway (tracheostomy or rare intubation)
- send blood for culture
- then :
- IV 3rd generation cephalosporin or chloramphenicol
- rifampicin to contacts
- **Recovery :** 2- 3 days
- **Prevention:** HIB vaccine

NOTE: : in x-ray

- croup : steeple sign (AP View)
- acute epiglottitis Thumb's sign (lateral View)



Thumb's sign (lateral

❖ Stridor :

- infectious : croup, bacterial tracheitis, acute epiglottitis
- mechanical : foreign body, external (hemangioma, goiter)
- since birth :

- laryngomalacia
- congenital subglottic stenosis

- others :

- hypocalcemia
- vocal cord paralysis

- if is stridor : - in less than 4 months age or continuous more than week = Direct laryngoscopy "i.e not croup"

❖ Whooping cough:

- peak : less than 4 months non vaccinated

- **stages :**

1- catarrhal "coryza" : 1_2 weeks
2- paroxysmal : 2_4 weeks

- spasmodic cough followed by inspiratory whoop (inspiration against closed glottis) worse at night.
- then post - tussive emesis.

3- convalescence : 1_2 weeks (recovery)

NOTE: most Contagious during early stage (2 weeks after onset of cough)

NOTE: in infant atypical presentation = neonate apnea

- young infants : CNS damage (hypoxia), secondary pneumonia

- **Diagnosis :**

- Pernasal swab :culture or PCR
- CBC : lymphocytosis

- **Treatment :**

- admission, isolation
- erythromycin

(in neonate use azithromycin) i.e erythromycin = pyloric stenosis.

- erythromycin : decrease symptoms in catarrhal stage but not paroxysmal.
- also give erythro. for contacts
- vaccinate if not vaccinated.

- **Complications:**

1- Pneumonia "by Bordetella itself or secondary bacteria" #1

2- Atelectasis : 2ry to mucosal plug
3- CNS damage (convulsions, encephalopathy)
4- Paroxysms:
- Subconjunctival hemorrhage
- Retinal hemorrhage
- S.C emphysema
- Hernia
- Rectal prolapse

◈ lower respiratory tract infection (LRTI) :

❖ Bronchiolitis :

- **Cause :**

- RSV 1#
- Human metapneumo virus 2#
- PIV, Influenza virus, etc.

- **Transmission :** Respiratory droplets

  #1 Serious respiratory infection in infants and more severe disease in :

  1- Cardiopulmonary disorder (CHD, bronchopulmonary dysplasia, cystic fibrosis)
  2- Neuromuscular disease
  3- Immunodeficiency

- **C/Of:**

  1- 1st phase : coryza
  2- 2nd phase : shortness of breath, poor feeding, noisy breathing, audible wheeze, low grade fever.

- **Examination:**

- Respiratory obstruction
- Hyperinflation of the chest (liver downward prominent sternum).
- Hyper resonance on percussion
- Palpable rhonchi
- On auscultation : Prolong expiratory phase Wheeze, fine crackles
- If severe : cyanosis, grunting.

Note : young infant may not have this classical presentation = apnea instead.

- **Investigation:**

- Most specific : antigen detection (ELISA or IF) or PCR (rapid).
- From nasopharyngeal secretions
- But you should do pulse oximetry to monitor oxygen saturation
- Chest x-ray : hyperinflation " increase lung translucency flat diaphragm".

- **Differential diagnosis :**

1. Asthma :

- later age of onset, no fever
- family history of asthma
- recurrence
- respond to bronchodilators
- increase eosinophiles

2. Cardiogenic asthma
3. GERD : frequent + emesis
4. Foreign body inhalation (focal area does not inflats or deflate).
5. Others:

– Organophosphate poisoning
– Bronchopneumonia with spasm

- **Treatment:**

- Supportive
  Monitoring (O2 saturation/apnea)
  O2 by nasal canula
  Fluids (IV or by NGT)
- If RF or apnea >> ventilatory support
- In the past >> Bronchodilators, steroids
- Admission if:
  ~ < 6 months
  ~ Hypoxia (saturation < 92% / PO2 < 60)
  ~ Unable to feed

- **Complications:**

1. Apnea: in young infants
2. Recurrence (but it's mild)
3. RF (Cause of death)
4. Dehydration (Cause of death)
5. Heart failure
6. Bronchopneumonia

- **Prognosis:**

- Most recover in 2 - 5 days, but there's risk of recurrent wheeze
- 50% of recurrent bronchiolitis >> develop asthma

- **Prevention:**

- Palivizumab injection for those who are prone to severe infection (CHD, YLBW, BPD)
  ➔ Monoclonal antibody

- **Other DDx:** FB inhalation

- Toddler: history of chocking, unilateral finding on examining (e.g. collapse, wheeze)
- CXR: Collapse
  - **Treatment:** removed by bronchoscopy

❖ Pneumonia:

- **Causative organisms:**

- Newborns >> GBS, E.coli, other gram -ve
- Infancy/young children >> viral
- Older children >> S.pneumonae, M.pneumonae, H.influenzae, Chlamydia pneumoniae

- **C/O:** Fever + RD: Increase RR (most important), active ala nasai, subcostal/intercostal recessions/grunting
  If severe >> cyanosis , Cough
- **Examination:**

- Consolidation (not usual, Crackles/wheeze)
- Also: Meningism, abdominal pain, shoulder pain (pleurisy)

- **Danger signs:**

1. Apnea
2. Cyanosis (Respiratory)
3. Convulsions
4. Impaired consciousness (CNS)

- **Diagnosis:** - CXR - CBC

RD = Respiratory distress

Nasopharyngeal secretions in young children >> Viral

- **Hospital admission in:**

1. Neonate
2. Danger signs
3. O2 saturation <92%
4. Inability to feed
5. Immunocompromised
6. Severe respiratory distress (e.g. granting)
7. Unsuitable home environment

- **Treatment**

✓ **General:**

- O2 by face mask or nasal canula
- Hydration
- Antipyratic
- Monitoring

✓ **Specific:**

➔ **< 1 year**

- In patient: (Neonate, infant + danger signs) → IV broad spectrum
  - Amoxicillin + Gentamycin + 3rd generation cephalosporin
  - You can add vancomycin if MRSA is suspected
- Our patient: Amoxicillin or Co-amoxiclav

➔ **> 1 year**

- In patient: Benzylpenicillin
- Our patient: Erythromycin

❖ Cystic fibrosis:

- Autosomal recessive (chromosome 7)
- Cystic transmembrane conduct regulator
- cAMP Cl-channel
  - **C/O:**

○ **Respiratory**

- Sinusitis, nasal polyp
- Recurrent infection, by abnormal organism e.g. P.aeruginosa
- Bronchectesis
- Increase risk of allergic bronchopulmonary Aspergellosis
- Hemoptysis

○ **Intestine**

- Infant >> Meconium ileus
- Older children >> Mucus plug >> Distal intestinal obstruction

○ **Pancreas**

• **Exocrine** >> Malabsorption

- Lipids: steatorrhea ~ Failure to thrive
- Proteins: hypoalbuminemia ~ Peripheral edema

  Dx: Decrease elastase in feces

• **Endocrine** >> DM #1 complication

- Biliary system: Cholestasis > Prolong neonatal jaundice
- Liver: Cirrhosis (later in life)
- Others: Male infertility >> Absence of vas deference & Rectal prolapse
  - **Investigations:**
- Screening >> Guthrie's test > immuno reactive (Trypsinogen)
- Confirm >> Sweat test Cl' >60 mmol/L Genetic testing

- **Management:** Multi disciplinary

☆ Respiratory ➔

  - Mucous: physiotherapy
- Mucolytic (Neubulized DNase Hypertonic saline)
  - Infections: Prophylactic >> Flucloxacillin
- IV antibiotics if developed infection

☆ Nutritional ➔

- High calories diet
- Pancreatic enzymes, replacement with meals
- Vitamin supplements

☆ Others ➔

- Treat complications as they appear
- Definitive management ➔ Bilateral sequential lung transplant
  N.B. Cause of death in CF >> Respiratory complications
- Electrolytes change in CF >>
- Decrease Na+
- Hypo Cl'
- Metabolic alkalosis

CF = Cystic fibrosis

2# Endocrine (DM, Thyroid)

❖ **DM:** Types

1- Type 1: Autoimmune (children)
2- Type 2: Family history, obesity (adult)
3- Type 3:
   - Neonatal (transient/permanent) 1 - 3 months
   - Secondary (CF/hemochromatosis/drugs e.g. steroids)
   - Genetic >> MODY : A.D

- MODY = Maturity onset diabetes of the young
- Mutation in the gene of Beta cells
- In < 25 years
- Genetic abnormality in insulin action
   - Mitochondrial DM

4- Type 4: Gestational diabetes

*** Type 1**

- C/O: Polyurea, polydepsia, polyphagia, weight loss, sometimes; secondary enuresis
- Or may present as DKA >>
  + Nausea, vomiting, abdominal pain, rapid deep breathing (Kussmal), decrease LOC

- **DDx of DKA:**

1- Acute abdomen
2- Gastroenteritis (no polyurea)
3- Dehydration
4- Pneumonia (rapid breathing)

- **Dx:**

- Sx + RBG > 200 (11. 1mmol/L)
- Fasting > 126 (7 mmol/L) (110 - 126 : impaired)
- 2 hours PP > 200 (140 - 200: impaired)

❖ **DKA: ▪Criteria:**

1- Hyperglycemia: > 200 (usually > 300) N.B. sometimes (Euglycemic DKA)
2- Ketonemia (>3) / Ketonuria
3- pH < 7.3 or HCO3 < 15 (Acidosis)

- **Causes:**

- Omiting the dose
- Inappropriate dose
- Stress (e.g. Infection)

N.B fully conscious patient can have DKA

- **Investigations:**

1- Blood glucose
2- Blood ketones
3- Arterial blood gases
4- U/E, creatinine
5- Urinary ketone/Sugar
6- Infection, CBC, blood film, etc

- **Management:**

- If comatose ~ NG tube and suction

1- Treatment of shock: 10-20 ml/Kg over 20-30 minutes

2- Treatment of dehydration: over 36 - 48 hours → 50% initial 12 hours . other next 24 hours
Deficit: 10% body weight (Litres)
Maintenance:
1st 10Kg × 100, 2nd 10 × 50, rest × 20
24 hours = 1/2 deficit + maintenance
Type of fluid:
Normal saline, or Ringer lactate until blood glucose is 250 then shift to >> 5% Dextrose + 1/2 normal saline (0.45 NS)

3- Correction of electrolytes:

- K+ (keep monitoring by ECG)
- **When to give?** When the child start to pass urine OR K+ < 5
- **How much?** 40 mmol/L (ampulla '20' in one drip)
- HCO3: Not given unless there is a severe acidosis , pH < 7 Causing circulatory failure
- **Why?** It will change to CO2 = cross the blood brain barrier, increase CNS acidosis and edema

4- Insulin therapy

- After 1 hour of fluid therapy
Type: regular insulin (short acting)
Dose: 0.1 U/Kg/hour by infusion pump → 0.3 U/Kg then 0.1 U/Kg/hour S.C
When to stop? When clearance of acidosis occurs(Normal pH/HCO3/ketones)

• **Aim of treatment?** 4-5 mmol (80-90 mg) decrease every hour

N.B: if blood glucose become < 100 or falls rapidly >> increase concentration of dextrose to 7.5%

- If blood glucose rise to > 300 → increase insulin infusion by 25%

- **Monitoring:**

- Vital signs hourly then 4-6 hourly after stable
- Blood glucose hourly

- Blood gases, ketones, U/E, 4-6 hourly
- Urine ketones, 2 hourly
- Renal chart
- ECG
- GCS
  - **After clearance of acidosis:**
- If this is at night (can't drink or eat) >> decrease infusion by 1/2
  = 0.05 U/Kg/hour (monitor 2 hourly)
- If in day >> give S.C insulin 30 minutes before stopping infusion
  - If known diabetic, return to usual dose
  - Unknown diabetic, 0.5-0.7 U/Kg/day (Regular 1/3 + NPH 2/3)

❖ Cerebral edema

▪Criteria:

- **Diagnosis:**

- Abnormal verbal/motor response to pain
- Decorticate or decerebrate posture
- CN 3,4,6 palsy
- Abnormal neurogenic respiratory pattern

- **Major:**

- Abnormal mentation
- Sustain decrease in heart rate, not due to impaired circulation or sleep
- Age inappropriate incontinence

- **Minor:** Vomiting, headache, lethargy, DBP > 95 , age < 5 years

- **Treatment:**

- 1st exclude hypoglycemia → Then → ICU admission → elevate head
- Give IV manitol (1g/Kg or 3 L N.S 5-10 ml/Kg over 20-30 minutes)
  + 1/2 hydration + intubation

  N.B after stabilization → CT or MRI (Normal CT doesn't exclude cerebral edema, it is C/O Dx)

  N.B Ketonemia + hyperglycemia and not DKA

1- If known DM → No admission

- Regular insulin (0.1 or 0.2 U/Kg or 10% usual dose)
- At lunch time + with usual dose
- Increase fluids (frequency but small volumes)
- Blood glucose/ urine analysis after 4 hours

2- For New case: Admission

- 0.5 U/Kg/day + (مخلوط)

- 10% of this as regular insulin as above or Sliding ? Scale → No need for IV fluids

❖ Thyroid:

- At birth: TSH surge (return in week)
- Preterm: normal TSH but low T3, T4 >> no need for thyroxin

Congenital hypothyroidism:

Why is it important?

- Common 1/4000
- Preventable
- Treatable cause of learning difficulties

- **Causes:**

1) Primary:
    - Thyroid Agenesis/maldescent "lingual" (No goiter)
    - Dyshormonogenetic goiter
      1st# in Sudan (perchlorate discharge test)
    - Iodine deficiency
2) Secondary: pituitary

- **C/O:**

- Course features
- Umbilical hernia
- Large tongue
- Double chin
- Low IQ (learning difficulties)

- **Screening:**

- Guthrie's test
- Others: CF, PKM

N.B Guthrie's test, depend on TSH detection
So >> may not detect secondary

- **Treatment:** Thyroxin after 2-3 weeks after birth
- **Prognosis:** good intellectual development

Juvenile hypothyroidism: autoimmune

Hyperthyroidism

Neonatal hyperthyroidism Antibodies from mother with Gravis

- **Treatment:** As normal (If not may be fatal) - Self limiting

Congenital hypothyroidism - Preventable cause of mental retardation

- **Causes:**

1) Agenesis, dysgenesis, ectopic thyroid (lingual)
2) Synthetic defects: TRH, TSH/ TSH receptors
3) Dyshormonogenesis
   Transport, organification, peroxidase deficiency, thyroglobulin defect, deiodine defects.
4) Maternal: autoimmune, transplacental, deiodinase defect transfer.
5) Use of radioactive therapy antenatally

N.B. 1ry .. very high TSH…2ry → pituitary.. panhypopitarism.. low TSH

- **C/O:**

- Neonatal screening F:M 2:1
- Normal birth weight and height.. increase HC
- Prolong neonatal jaundice (every thing is slow)
- Sluggishness, resp. Difficulty , sleep a lot. Cry a little, constipation, umbilical hernia, hypothermia.
  * Progress to physical /mental retardation by 3-6months
  * Wide fontanelle
  * Delay puberty.

✓ TSH > 40ml/L → prechlorate discharge test.. dyshoemonogenisis.

- **Treatment:**

- Sodium L. thyroxin
- **SE**: pseudotumor cerebri

✓ Neonate: 5-10 microg/kg/day
✓ Children: 4
✓ Adult:2

❖ **HypoCa+2 :** (if congenital: neonates)

- PTH def.: Aplasia, hypoplasia, DiGeorge syndrome. 22q11 deletion.
- 10P13deletion, maternal DM, retardation and dysmorphism. (sanjad sakal)AR .
- Maternal hypo parathyroidism
- Genetic defects
- Autoimmune parathyroiditis → Isolated, type 1 autoimmune, poly endocrinopathy .
- Infiltration, hemosclerosis, Wilson. - Mg+2 def. - Vit D def. etc.

- **C/O:**

- Mild: asymptomatic
- Muscle painful cramps
- Numbness and tingiling (in hands and face) +ve. Signs

- **Lab:**

- n = 9-12 here decrease
- Increase PO4 -3
- Decrease ionized Ca+2
- Decrease or normal ALP
- Prolong QT interval.

- **Treatment:**

- IV 5-10ml or 1-3 mg/kg 10% solution of ca2+ gluconate.
- Monitor HR
- Vit D
- Increase Ca+2 and PO4-2 intake

  SE of Ca+2 therapy: abdominal pain, constipation, polyuria.

  N.B. DiGeorge syndrome: congenital def. In PTH or with:
- Thymic aplasia, defective immunity, cardiac defects, facial abnormality
- In older children: hypo PTH is usually an autoimmune with Addison disease.

  N.B. In severe hypoglycemia at home: jam /sugar
- At hospital: 10%dextrose 4m.

3# Short stature

- **Definition** < 3rd centile for appropriate age/sex. <2 SD .
  N.B. pathological usually <3.5 SD
- Growth velocity: (cm/year) <25th percentile (abnormal)
- **Causes:**

1) Non pathological:	2) Pathological:
• **Familial:** - **Genetic, with in centiles for mid parenteral ht.** "Mid parenteral ht": - **Male: (father ht +mother ht /2) + 6.5** - **Female: (father ht +mother ht /2) - 6.5** **- 2 SD. Subtract 10cm** **+2 SD add 10 cm** - **Bone age =chronological age** - **Normal onset of puberty** • **Constitutional delay:** - **Delayed puberty, delayed bone age in otherwise normal boy → Constitutional** - **They will achieve growth late** - **Family history is usually positive** • Treatment: **reassurance** - **Some need sex hormones** - **Kick off dose at age15** ➔ **Short female do karyotyping to exclude Turner.**	I. Nutritional/chronic illness (celiac, crohn's, CRF, CHD, esp major) Decrease wt. delayed bone age. II. Psychological"maternal deprivation syn." III. Endocrinological: Increase wt, delayed bone age 1. Hypothyroidism 2. Growth hormone def.(<10mg/ml but if more does not exclude it) or resistance (Laron's synd) & Insulin like growth factor def. 3. Excess steroids .. cushing - CAH congenital adrenal hyperplasia.. they may start as tall but when reaching puberty end up as short adult. 4. Rickets., hypo parathyroidism IV. Syndromes: Turner, Downs, nooran's, silver russe, broader wili (hypotonia, hyperphagia obesity, developmental delay) V. Skeletal abnormalities: (disproportionate short stature) Confirmed by measuring. → Sitting ht: base of spine to the top of the head. Subischial leg length Total ht

- Achondroplasia.. legs > back
- Hydrocephalus. Recurrent otitis media. Dental malocclusion
- Dx: lumbar x-ray (L1-L5 packing) & Genetic study.
- Mucopolysacaridosis .. back > leg
- All are AR except: Hunter x- linked +no corneal cloudiness. (MPS2)

❖ Severe short stature:

- Laron's syndrome
- Idiopathic short stature

- Primordial dwarfism
- SHOX gene abnormalities (short stature homeobox) → on chromosome X

  N.B. SHOX gene.. present in both copies → severe short stature.
- Absence of one copy → short stature in Turner
- Additional copies →tall stature in klinfelter syn.

  N.B. G.H. indications..

1) GH def.
2) Chronic renal failure
3) Turner syndrome.
4) Prader willi
5) SHOX def.



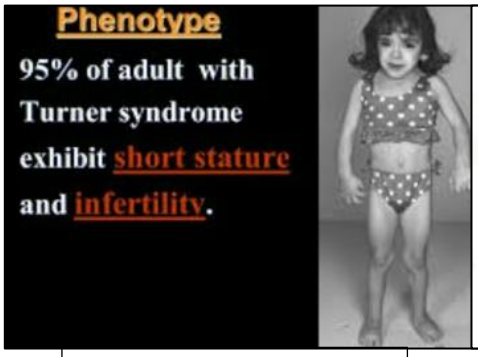
Prader willi syndrome



Laron's syndrome



Kallman syndrome



Turner syndrome



Down syndrome

❖ Tall stature > 97th centile

- **Causes:**

1- Familial #1
2- Obesity.
3- Secondary... increase growth hormone (gigantism)
- Increase sex hormone(e.g. CAH)
- Increase thyroid hormone
4- Syndrome.. klinfelter, marfan, homocystinuria



Marfan's

***Marfan's..**

- Tall, arm span > length, arachnodactily.
- High arch palate, ectopia lentils (upward) ,
- Joint elasticity, cardiac (AR , MVP, , aortic dissection)

***Homocystinuria..**

- = Marfan VS
- Ectopia lentis (downward)
- Joint stiffness
- + Mental retardation
- Thromboembolic phenomena.

4# Puberty..

- Adrenarche.. DHEA (before 2yr)
- Gonadarche.. GnRH, FSH/LH, sex hormone... secondary sexual characteristics
- Growth spurt.. DHEA,sex hormone, GH

- **Definition :** Physical / psychological changes that turn child into adult capable of reproduction.

# Female:	# Male:
- **Start8-12 year (average11yr)** - **First sign.. breast bud (tunner2)** - **NB.. T4: breast on breast.. start menst.** - **Pupic hair+growth spurt** - **Then after 2.5-3 yes.. menarche** - **T2 Growth spurt: immediately & short Duration.** - **Duration: 2-3 yrs** - **Caudo-cephalic... gynecoid pelvis**	- Start 9-13 year (average12yr) - First sign: testicular enlargement >_ 4ml (T2) by orchidometer - Pubic hair - Then after18months from the start.. growth spurt - Growth spurt.. late.. long duration (T4, T5) - Duration 3-5yrs - Android.. wide chest - Boys are longer by 13cm

- **Tuner staging:**
- 5 grades No. T0, T1....T5

N.B. At time of puberty.. is close to bone age.. not chronological.

❖ Puberty disorders..

- **Perciocus puberty** (pp):

- < 8yr .. female
- < 9yr .. male
- F>M

- **TYPES:**
- Central (true) : GnRH dependant.. high FSH and LH
- Psudo (false) ; GnRH independant.. low FSH and LH

1. **Central:**

- Idiopathic (familial) (constitutional)
- CNS insult (eg. Infection, congenital toxoplasmosis, encephalitis, meningitis) Tumors (eg. N.F), radiation, surgery, trauma, hematoma, hydrocephalus,
- Hypothyroidism.. only cause with delayed bone age.

2. Psudo:

- Mccwne.. albright syndrome
- Adrenal.. CAH , tumors.
- Testicular tumors (eg. Leydwigs cell tumors)
- Ovarian tumor
- Hepatoplastoma (hcG)
- Exogenous steroids
- **N.B.** independent peripheral puberty may change into true
- Hypothalamus read bone age.

- **Female:**

- Usually idiopathic(85-95%)
- Pelvic U/S. multicysts in ovaries, Enlarged uterus

- **Males:**

- Usually organic

  Testis ex:
- Enlarged-- tumors
- Enlarged -- central pp
- Small -- adrenal
- **N.B.** peripheral pp...no testicular enlargement.

  # Mccwne -Albright syndrome..
- Disturbance in signaling pathway by G protein.
- Ovaries.. pp
- Bone.. polycystic fibrous dysplasia
- Skin.. café' eu lait

- **Management of pp:**

- Central.. gonadotrophin analogue (slow releasing) (LH RH analogue)
- Peripheral..
- Medroxy progesterone acetate.. inhibit ovarian / Testis steroidegenesis.
  Or ketoconazole Or testolactone (diuretics) .. aromatase inhibitor
- Psychological/ behavioral support

  When to stop? About10 years.

- **Premature thelarche:**

- No other signs of puberty (no vaginal bleeding, no cx secretions, no axillary / pubic hair, thickening, no growth spurt, no advanced bone age.)
- No investigation needed(bone age can be done)
- Tx: reassurance +follow up.
- Age: 6 months- 2 yes (make sure there is no exogenous steroids)

- **Premature puberache adrenarche:**
- Rule out late onset CAH , ovarian tumors and adrenal tumors.
- **Treatment:** reassurance. Follow up
- **Risk..** increase risk of PCOS
- **Premature menarche**.. mostly regress spontaneously

Delayed puberty..

- > 15yrs .. female
- > 14 yrs. male
- (No signs of puberty)
- Unlike pp. M>F
- **Cause** is constitunal delay in growth /puberty.
- Problems.. inadequate physical growth.. osteoporosis.. psychological.
- **Other causes:**

Hypogonadotrophic hypogonadism.	Hyper gonadotrophic hypogonadism.
1- Wt. Loss (anorexia nervosa, malnutrition, chronic illness, eg. Celiac disease) **2- Athelets (excess physical training)** **3- Isolated gonadotrophin def. (Eg. Kallman's syn. +anosmia)** **4- CNS tumors eg. Craniopharyngioma** **5- Hypothyroidism #1- idiopathic**	- Infertile - Testis failure... Kleinfilter syn. (47XXY , upper segment <lower segment) - Ovarian failure.. Turner's syn . - Acquired gonadal dis. " radiation, torsion " - CAH

Syndromes.. Prader willi **&** Bardet peidel (extra digits)

- **Treatment:**
- Oral oxandrolene "weak androgen "can be used in young males. ... just accelerate growth.
- In older males.. IM testosterone or Patches. low dose.
- Growth secondary sexual char.
- Female.... estradiol first then add progesterone.

❖ **Disorders of sexual differnation (Ambigous Genitalia) :**

1) Viriliaztion of females #1
   &CAH
   -> female pseudo hermaphroditism

2) Inadequate androgens in males
   (male pseudo hermaphrodite)
   - Androgen Insensitivity Syndrome
   - 5 Alpha reductase deficiency
3) Low Gonadotropin (eg. Prader Willi)
4) Ova/test disorders of sex differentiation

❖ CAH

- AR
- 90% -> 21 alpha hydroxylase deficiency

- **C/O :**

  Females: Virilaization at birth → enlarged (hypertrophied) clitoris
  Fusion of Labia majora

  Males: Enlarged penis, scrotal pigmentation at birth
  - 80% salt losing crisis (shock/Vomiting)
  - 20% Tall stature , P.P
- **Investigations:**
  - Decreased Na+ , Increased K+ (ie. Low aldosterone)
  - Metabolic acidosis
  - Hypoglycemia
  - High urinary 17 hydroxy progesterone
- **Treatment:**

- Glucocorticoids (i.e. decrease ACTH level -> decrease testosterone and allow normal growth and maturation)
- Flurocortisone (mineral corticoids) -> if there's salt losing
  N.B. infants (before weaning) may need added NaCl
- Increase Na+ in food
- Follow up: Growth & Plasma 17 alpha hydroxyl progesterone
- Increase Doses in stress (illness,surgery)
- Surgery for females to correct external genitalia abnormality

- **Acute Presentation:**
  1. Hydrocortisone
  2. Flurocortisone
  3. IV fluids : NS (Na+ + 5-10% D glucose)
     IV) 25 mg for newborn and < 3yrs
     500 mg for 3-12 yrs
     100 mg for > 12 yrs

5# **Oncology:**

Leukemia:

- CBC:
- low Hb (indication of chronicity),
- Low platelets,
- Normal , increased or decreased WBCs.
- P.B.P: blasts
- Bone marrow aspiration: >25% blasts.

How to differentiate between ALL, AML?

1- Morphology in blood smear or B.M aspirate. (myoblasts have auer rods)
2- Cytochemical stain:
- ALL: +ve PAS
- If T.ALL: +ve acid phosphatase
- AML: +ve Sudan black, +ve myeloperoxidase.
3- Immunophenotyping (flow cytometry):
- T.ALL : CD3
4- Cytogenic studies: prognosis (most important factor in AML)
- 1/3 of cases: 47 chromosomes (hyperdiploid)
- **Tanslocation in:**
- T(12,21): #1 in ALL (good prognosis)
- T(9,22): phladelphia (poor prognosis)
- T(4,11): poor prognosis , usually in infants, 2ry AML.
- L.P: CSF analysis (mets)
- CXR: mediastinal mass (in 50% of T.ALL)
- Serum K+, Ca2+,urea, Po4-2, uric acid for tumor lysis syndrome.
- Increased uric acid, K+, urea, Po4-2 and decreased Ca2+=> after starting chemotherapy.

- **Prevention:** hydration, allopurinol.
- **Prognostic features in ALL:**
  1. Age: 1-10 years : good
  2. Sex: female : good (not accepted now)
  3. Class: L1; common ALL: good L3; TALL: bad.
  4. CNS tx: if abscent: good.
  5. DNA index: >1-16 : good?
  6. Cytogenetics: t(12,21): good

- T(9,22) t(4,11): bad

7. Response to therapy:

- Early < 7 days: good
- WBCs < 1000 good

8. WBCs at presentation > 20,000 : bad

N.B:

	CML	Leukmoid reaction
WBCs	Increased	Increased up to 50,000
Philadelphia	+ve	-ve
Basophil/acidophils	Increased	-
WBCs ALP	-ve	+ve

- **Treatment:**

- Supportive. (RBCs, plts, infections)
- Prevention of tumor lysis syndrome.
- Chemotherapy:

  Induction 4 weeks

  Consolidation (CNS tx)

  Continuation : 2 years (3 years in boys)
- B.M transplantation.

❖ **In ALL:**

1- Induction :

- IV vincristine/week
- Prednisolone: throughout induction
- I.M L.asparaginase And for high risk + Daunorubicin I.V

- CNS rx , if +ve sx. Or T.ALL : CNS irradiation (radiotherapy)
- Other wise: intrathecal methotrexate.

2- Consolidation (for residual leukemia):

- High risk: cyclophosphamide, cytarabine
- Low risk: 6-mercaptopurine (6-MP)

- **Maintenance:** at least 2 years (3 years in boys)

- 6-MP daily, oral methotrexate weekly
- IV vincristine/ 4 weeks + 5 days of oral dexamethasone
- 12 weekly IT methotrexate

3- Re induction / delayed intensification

N.B. AML is more common with down than ALL.

❖ Lymphoma:

✓ 3rd most common malignancy in childhood after leukemia and CNS tumors.

✓ Types:

- Hodgkin's lymphoma HL = 40%
- Non-hodgkin lymphoma NHL = 60% (#1 in AIDS pts)

❖ NHL:

- 3 histological types:

1) small (non cleared cells): Burrkit's lymphoma ; #1 in Sudan.

- Mature B cells.
- Sporadic: abdomen
- Endemic: head and neck (jaw)

2) lymphoblastic:

- T- cells or rarely pre B cells
- Neck, Anterior mediastinum.

3) large cells.

- T or B cells .
- LN, skin, soft tissue, bone.

❖ Burkitt's lymphoma translocation:

- C-myc oncogene (chromosome 8)
- Gene 2,14,22
- T(2,8) , t(8,14) , t(8,22)

- **Epidemiology:**

- HL : Bimodal (adolescence , >50 years)
- NHL: increase with age, in immunodeficiency, #1 tumor in AIDS

- **C/O:**

1. # HL:

- Painless, firm rubbery , non-marked lymphadenopathy
- Usually cervical, supra clavicular
- +/- B symptoms: night sweating, weight loss >10% in 6 months, fever >38

2. # NHL:

Burrkit's:

- Sporadic: abdominal pain, DDx for acute appendicitis
- Endemic: jaw tumor

Lymphoblastic:

- As acute leukemia or superior vena cava syndrome
- Shortness of breath, cough . **Large cells:** B.sx

N.B: B. sx = HL **&** Extranodal = NHL

- **Investigations:**

- CBC: low Hb = poor prognosis;
- Anemia of chronic disease , autoimmune hemolytic anemia
- ESR: increased >100
- Electrolytes : uric acid.
- Tissue Dx: eg LN biopsy 'not aspirate'

- **Staging:**

- CXR, CT chest,abdomen, pelvis.
- PET scan
- BM aspirate.

  N.B: Hallmark of HL = Reed-strenberg cells (RS cells)

Histological types of HL:

- Nodular sclerosis (60%): mononuclear RS cells → (lacunar cells).
- Mixed cellularity
- lymphocytes predominant : best prognosis
- Lymphocytes depleted: worst prognosis

Ann-Arber staging of HL:

1) one LN group
2) >,= 2 on the same side of diaphragm
3) >,=2 on both sides
4) Mets (liver, BM,..)

- A: no B.sx
- B: +ve B.sx

- **Treatment:**

  Chemotherapy +,- radiotherapy

- **Side-effects:**

✓ Early: nausea, vomiting, alopecia
✓ Late:

- Vincristine: peripheral neuropathy
- Adinicmycin "doxorubicin": cardiotoxicity
- Bleomycin: lung fibrosis
- Cisplatin: ototoxicity, nephrotoxicicty.

- **Prognosis:** Good

N.B. HL : immunological disorders (SLE, RA, a-telangictasia, HIV, EBV, agammaglobulinemia).

Neuroblastoma :-

- Primitive neural crest cell:
- Adrenal medulla
- Sympathatic ganglia
- **Epidemiology**:
- # 1 malignancy in infancy
- # 1 extracranial childhood tumor of solid organs.
- 1-2% +ve family hx .
- ✓ Infants : cervical / thoracic
- ✓ Children : abdomen
- **C/O :-**
- Very wide spectrum :
- Spontanous regression __> rapid progression and mets .
- **Sites :**
- Most common: -Abdomen75% -Adrenal medulla -Retroperotenial sympathatic ganglion.
  Others : chest then pelvis then neck.
  N.B. : < 1 yrs → abdominal mass.
- **>** 1yr → mets usually bone.

#1 _presentation is bone pain and limp. Most important prognostic factor is age.

- **Symptoms :-**

- Abdominal mass " hard_ smooth_ non tender" maybe crossing midline.
- Neurological : . spinal cord compression " paravertebral site".
  . Hornor's syndrome.
- Paraneoplastic : watery diarrhea _ perfuse sweating
- Dancing eye , dancing feet .
- HTN "not common" due to compression of renal artery.

N.B. : no increase in catecholamines.

- **Mets :** bone VID , liver .

N.B. : mets to orbital wall : periorbital ecchymosis.

- Unique : stage 4S : favourable prognosis samll primary tumor with limited mets to skin, liver , bone marrow.

- **Investigations :-**

- CBC
- Plain imaging : calcification
    - Incentric → good prognosis .
- U/S or CT : to differentiate b/w wilm's tumor and neuroblastoma .
- Urinary VMA : homovanillic acid screening.
- Tissue Dx :
    - Cytogenic studies : N-myc or MYCN amplification → rapid progression → poor prognosis.
- 11q deletion , poor prognosis.

- **Staging:-** CT , bilateral bone biopsy / aspirate / MIBG scan ("metaiodo benzyl guanidine " → neurotransmitter accumulate in catocholaminergic cells).

1. Confined to organ
2. Not confined , but don't cross midline.
3. Cross midline.
4. Mets. " remember 4S"

N.B.: Blue barry muffin in neonate → skin mets.

- **Treatment:**
    - Localized : surgical excision.
    - Mets : chemotherapy , delayed surgical excision
      Radiotherapy : retinoic acid.

In some cases → high dose chemo + peripheral blood stem cell rescue .

#Wilm's tumor :- Malignant precursor : metanephric blastoma cell.

- **Epidemiology :-** #1 Renal malignancy in children. Age: 3 – 3.5 yrs
- **Associations:-**

A) WAGR syndrome :
   - (Chromosome 11) → WT-1 gene
   - Wilm's, Aniridia, Genitourinary abnormality (
   - Hypospadius , horseshoe kidney) , mental retardation.

B) Beckwith wiedwmann syndrome BWS :
   - Hemihypertrophy, macroglossia, omphalocele, transient hypoglycemia .
   - On microscopy → adrenal cytomegaly .

C) Denys-Drash syndrome.

A & b screen by abdominal U/S

- **C/O:** Asymmetrical abd.mass, hematuria, HTN
- **Investigations:** 1st : US Best: CT
- Calcification in peripheries: → CXR , CT chest → lung mets
- Histopathology → confirmation of disease (essential) .
- **Staging:**

1) 1 : within capsule.
2) 2 : penetrate capsule.
3) 3 : non hematogenous extension (IVC, LN).
4) 4 : hematogenous mets (eg.lungs VID).
5) 5 : bilateral kidney involvement.

- **Treatment:**
  - Nephrectomy, chemotherapy, radiotherapy (advanced)
  - Prognosis: good
  - Histology is the most important factor.

CNS Tumors:-

VS in adults → here mainly 1ry

- # 1 solid organ malignancy in childhood.
- # 1 mortality cause of malignancy , usually < 7yrs.
- **Some types :-**

Astrocytoma : (glioblastoma multiform).

- In the cortex (seizures , focal neurological signs , hemiparesis).
- Increased ICP (headache , morning vomiting , visual problems , 6th cranial nerve palsy , papilledema , in infants : increased HC).

✓ Supratentorial → choroid plexus papilloma .
✓ infratentorial .

Craniopharyngioma :- Reminants of rathke pouch .

- **C/O:** Increase ICP + visual problem (bitemporal heminopia) + pituitary dysfunction (growth failure / DI / increase wt) Post op → DI

Medulloblastoma (cerebelloma) :- Increase ICP + C.ataxia + eye signs

Also : ependymoma . → Seed through CSF : mets in spine.

Brain stem glioma :

- No increase ICP + CNS palsies + pyramidal sx "eg. weekness " .
- Very poor prognosis : only palliative tx .
- Excision is impossible .
- **Investigations:** Of choice : MRI . CSF for histopathology .
- **N.B.:** most common : Astrocytoma .
- (spectrum from bening to highly malignant → glioblastoma multiforme).
  2nd is medulloblastoma .

Retinoblastoma :-

- 5% of visual impairment in childhood.
- Unilateral or bilateral .
- Hereditary form (AD – chromosome 13).
- All bilateral and 20% unilateral.
- **C/O :-**
- White reflex (normal eye : red reflex).
- **DDX :** cataract , ratinopathy of prematurity , chorioretinitis , viterous hemorrhage .
- Squint.
- During screening in family hx pts.
- **Investigations:**
  - MRI
  - Examination under anasthesis EUA
- **Treatment:**
- Early : chemo
- Late : enucleation of eye +/- radio .
- Recurrance : radio

N.B. : pts with inherited retinoblastoma are at risk of other tumors eg : osteosarcoma.

Sarcomas :-

- Soft tissues :-
  Rhabdomyosacoma.
  Fibrosarcoma.
  ....etc.
- Bones :
  Osteosarcoma
  Ewing sarcoma.

Increase risk :

- Some syndromes (NF , Li-fraumeni , BWS).
- Hereditary retinoblastoma and osteosarcoma .

- **C/O :-**

  # Rhabdomyosarcoma 2- 6 or 15 - 19 yrs

  According to site:-

  - 40% head/neck : periorbital swelling, proptosis ,nasal bloody discharge or obstruction, chronic ear infection,neck mass,.....etc
  - 20% genitourinary: dysuria, hematuria, urinary obs sx,...
  - Scrotal swelling
  - Blood stained vaginal discharge
  - trunk/ extremeties

  # Osteosarcoma : Distal femur or proximal tibia or proximal humerus

- persistant local pain preceed the mass

- **DDx**: hx of trauma
  - Lytic lesion with calcification on x-ray

    # Ewing sarcomas

    - Pelvis, skull, femur, sternum .. Most common sites (pelvis, femur)
    - Pain, limb, mass, sys sx
  - **DDx**: osteomyelitis
  - **Investigations:-** Imaging (xray , MRI) & biopsy

N.B.: rhabdomyosarcoma

- Embryonic (young child , head/neck, GU)
- Alveolar (older , extremeties)

- **Treatment:**

- Rhabdomyosarcoma : combination therapy(surgery, radio,chemo)
- Bones: neoadjuvant chemo..surgery..post.operative chemo
- Limb salvage or amputation

  N.B.:- Osteosarcoma: not radio sensitive unlike ewing's

- Good prognostic features of rhabdomyosarcoma:-
  - Embryonal histology
  - Head and neck
  - Localized
  - Small (less than 5 cm)

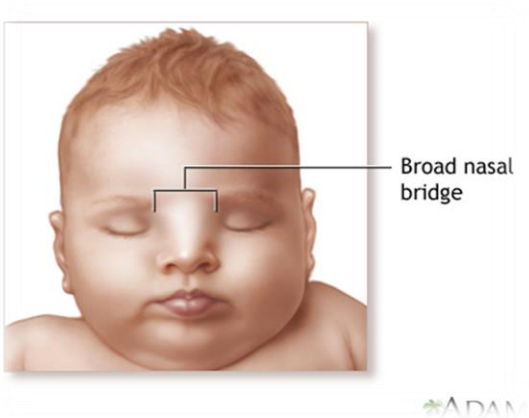
6# Renal

- GFR at term: 15-50 ml/min/ SA (low)
- Adult :80-120 ml/min/SA
- •preterm is even lower.
- *congenital abnormalities (detected by antenatal US)

•Renal a-genesis :-

***Bilateral : (potter's syndrome)**

- Oligohydramnios
- Potter's face: flat, low set ears, broken nose, epicanthic fold, down slant of eyes.
- Limb defects (Talipes, quinavarus, club foot)
- Pulmonary hypoplasia
- Outcome: still birth or die shortly after birth(i.e Respiratory faliure)



*** Unilateral :**

- VACTREL, infant of diabetic mother, Turners, poland.
- VACTREL: Vertebral abnormalities, Anal atresia, Cardiac abnormalities, TOF, Esophageal atresia, Renal agenesis, limb hypoplasia.

• Multicystic Dysplastic Kidney :-

- Failure of fusion between ureteric bud -> nephrogenic mesenchyme .
- Pathogenesis: non functioning, just fluid filled cystic kidney.
- Outcome: involution at 2yrs
- Nephrectomy if large, HTN
- (bilateral = potter's syndrome)

• **PCKD :-** #1 inherited kidney disease AD PCKD

<u>- other causes of large kidney-></u>

- ^ Tuberus sclerosis
- ^ Bardet-biedl syndrome
- ^ Von hippel lindaw -> Renal cyst, hemangioblastoma (cerebral/ retina), phenochromocytoma

- **Bardet-Beil syndrome :-** Obesity + polydactly +GU abnormalities (Renal/hypogonadism) neuro-optic (mental retardation, retinitis pigmentosa)
- **Abnormal migration :-** Pelvic kidney, horse shoe kidney (below inferior mesenteric artery) turner's
    - Risks : infection/ obstruction.
- **Duplex syndrome :-** (Female never gets dry)
    - Premature division of ureteric bud
    - Ureter from upper pole: un usual sites -> Vagina, urethra, urotrocele (prolapse into the bladder)
    - Lower: Reflux.

• **Bladder Extrophy :-** Failure of fusion of the infra-umblical midline structures.

• **Prune-Belly syndrome** "absent musculature" :- Abscence of anterior abdomenal wall muscles + Mega Cystis, Mega ureter, crypochoridism

• Obstruction of urinary flow :-

- Pelvic ureteric junction -> unilateral hydronephrosis.
- Bladder neck obstruction -> PUV (male) -> Bilateral.

Consequences :-

- Hydroureter, hydronephrosis
- Thickening of the bladder wall + diverticula
- Renal dysplasia "small/ non functioning"
- Severe bilateral = Potter's syndrome

How to deal with anomaly Dx antenatally?

- if bilateral hydronephrosis + dilated lower urinary tract —>
    - In male = U/S within 48 hrs —> PUV.
- if unilateral in male or bilateral in females —> U/S in 4-6 weeks.
- in all 1st give prophylactic antibiotics.
- If U/S is normal —> stop Antibiotics and repeat U/S in 2-3 months.

❖ **PUV : Post Urethral Valve**

Definitive Dx : MCUG (A micturating cystourethrogram (MCUG) is a scan that shows how well your child's bladder works. It is used to diagnose why your child may have urinary tract infections. It is also used to show up any abnormalities with your child's urinary system).

- **Treatment :** urgent surgery (cystoscopic ablation)

❖ VUR : Vesicoureteral reflux

- retrograde flow of urine from the bladder to ureter and kidney.
- **Causes :** Primary: incompetence of VUJ —> congenital —> resolve alone in childhood Structural —> remain
- Secondary: to obstruction or urinary tract anomalies (PUV, neuropathic bladder "meningomyelocele" , duplication of ureter "lower"). its familial
- **Risks:**
- Recurrent UTI —> scarring (HTN, CDK) .
  (Incomplete emptying of the bladder —> intrarenal urine)
- **Dx:** MCUG
- **Grading:**
  1: if reflex into ureter
  2: ureter + calyces (no dilatation).
  1&2 usually revolves alone, low risk of scaring .
  3: mild/ moderate dilation.
  4: gross dilatation of ureter and calyces.
  5: gross dilatation, tortuosity, loss of papillary impression.
  4&5 —> need surgery usually, high risk of scarring.
- **Treatment of VUR :-**
  - Antibiotics Cotrimoxazole, nitrofurantoin prophylaxis.
  - Surgery: controversial indications.

❖ UTI:

- **Epidemiology:**
  - Infant : M > F (PUV)
  - After that : F > M
- **Types:**
  - Cystitis (no fever)
  - Pyelonephritis (+ve fever)
- **Why Is it important?!**
  1- 50% are due to urinary tract anomaly.
  2- may cause renal scaring (HTN , CKD).
- **C/O :-**
  - infant : no specific Sx
    (Remember: jaundice/ sepsis/febrile convulsion)
  - classical Sx appears late

N.B Dysuria DDx : cystitis/ Balanitis "Balanitis is an inflammation of the glans, or the head, of the penis" / vulvitis .

- **Diagnosis :-**

- Collection of samples :- • children in nappies —>

  - stream catch urine.
  - adhesive plastic bag.
  - catheter / suprapubic aspiration >if there is urgently (eg; very sick)

• older children —> Mid stream urine MSU.

- What to do with Urine?!

1. Urine dipstick (nitrate / leukoesterase).
2. Microscopy.
3. Culture (not done if 1,2 are negative).

To diagnose —>

1. Growing of > 15 CFU/ml of single organism OR
2. Single organism /ml if SPA sample

N. B any infant with unexplained fever >38 → urine sample for analysis

• **Causative organism :** #1 E.coli (>90%)

- Others: Klebsiella, proteus, (high in boys, found under prepuce, high PO4 stones —> by alkalizing urine), pseudomonas : underlying structural abnormalities.

• **Predisposing factors for UTI:**

1. Antenatally Dx anomaly in urinary tract.
2. Incomplete bladder emptying:
   - infrequent voiding
   - incomplete micturation
   - PUV
   - VUR
   - neuropathic bladder
   - constipation (by loaded rectum)

Other investigations:- Some say : for any 1st attack

- Some say if :

1. < 6 months
2. Recurrent
3. Atypical —>
   - severely ill, septicemia
   - failure of Treatment in 48 hrs.
   - non E coli
   - high creatinine
   - abdomenal or renal mass.

U/S

- < 1 year —> MCUG / DMSA
- 1-3 years —> DMSA

  N.B. functional investigation (eg DMSA) are done after 2 months of UTIs —> i.e. to avoid false +ve. Not to miss new scaring from current attack.

- **Treatment:**
  - ✓ Infants :
  - < 3 months —> I/V 3ed GC
  - > 3 months —> oral Amoxicillin / Coamoxicalv (7-10 days)
  - ✓ **Children:**
  - Cystitis —> oral Antibiotics for 3 days
  - Pyelonephritis —> oral antibiotics for 7 -10 days

N.B. prophylactic Antibiotics: (trimethoprim /nitrofurantoin) ->

1. Severe VUR
2. UT anomaly + age < 2yrs
3. Upper UTI
   - (avoid amoxicillin ie poorly absorbed)

Nephrotic Syndrome

Def:

1. Massive proteinuria (> 40mg/m2/hr —> >2g/day)
   Or urinary protein = urinary creatinine
   Up : UCr > 200 mg/mmol
   (Nephrotic range)
2. Hypoproteinemia
   (protein < 5.5 g/dl / albumin < 2.5 g/dl)
3. G. Oedema
4. Hyperlipidemia

DDx for proteinurea :

1. Transient proteinurea, eg: severe exercise, seizers, dehydration, fever ...
2. Orthostatic (positional) proteinurea —> tall, thin

- #1 cause persistent proteinurea in school age , adolescents
- Upr : UCr > 0.2 in morning samples exclude it

3. Tubular(only low molecular weight proteins) —> ATN, pyelonephritis ,Fanconi syndrome (glucoses urea + electrolyte loss)

Fanconi syndrome is a rare disorder of kidney tubule function that results in excess amounts of glucose, bicarbonate, phosphates (phosphorus salts), uric acid, potassium, and certain amino acids being excreted in the urine.

4. Glomerular —> eg post streptococcus GN, RPGN, ...

- (LMWP + HMWP) + evidence of glomerular diseases (eg RBCs cast)
- LMWP : low molecular weight proteins
- HMWP : High molecular weight proteins

- **C/O :** Usual presentation:
  - **G**eneralized edema(starts periorbitaly)..all body including (effusions,ascites, scrotal swelling).

Complications of nephrotic syndrome:-

- Infections

Causes:- loss of IG / complement

- fluid edema is a good medium
- hypovolemia..less splenic perfusion
- Most common: URTIs

Others:-

- Spontanous peritonitis by strept.pneumonae
- T.B
- cellulitis
- severe chickenpox

- Thrombosis

Causes:- loss of antithrombin 111

- more fibrinogen synthesis
- hyperviscosity
- thrombocytosis (esp. By steroids)

- Hypovolemia

C/o : abdominal pain/ fainting

What to do? I.V albumin

- # avoid crystalloids

Diagnosis :-

- Urine dipstick...proteins > or = ++
- Urine protein... 24 hrs urinary protein Excretion (the gold standard)
- Serum Protein, cholesterol, complement
- Electrolytes/RFT ... should be normal

- **Typical N.S :-**

- age: 2- 8 yrs
- no gross hematuria (microscopic in 25 L of ???)
- no HTN
- no renal impairment
- normal complement
- no systemic disease
- responsive to steroids

- **Types of nephrotic syndrome:-**

1. Primary :(most common, minimal change)
2. Secondary : malignancy,drugs, infections..etc
3. Congenital

- < 3mons, AR.
- proteinuria is severe(cause of death)
- nephrectomy+dialysis
- Then transpantation when grows older.
- has high mortality and associated with TORCH.

4. Infantile (4_ 12 months)

- **Treatment :-**

- Specific : steroids.
- 60mg / m^2/day = 2mg/kg/day
- Divided into 2 doses per day for 4 weeks
- Then:
- 40mg/ m^2/day = 15mg/kg/day
- 1 dose alternative day for 4 weeksThen:
- 25 mg/ m^2 / day for 2 wks. (alternative days).

Then →

- 10mg/m2/ day for 2wks (alternative days).
  *all 12 wks duration (3 M)

Treatment / prevention of complication :

1) Infection :

- Vaccination for pneumococcal / H. Influenzae
- Aggressive ttt of infection.

If chicken box → acyclovair

2) Thrombosis : mobilization.
3) Edema : salt restriction (not fluid)

- If failed → loop diuretics
- If failed → IV 25% albumin + above measures.

- response to Tx :
  - No proteinuria for 3 consequent days.
- steroid sensitive : usually this response occurs
  - In 7_10 days (2wks).
- steroid resistant : No response →
- Proteinuria >++ after 4 wks of ttt. (Some says 8).
- What to do ?
- Decrease dose(40mg/m2/day) + add cytotoxic drugs
- Send for biopsy.
  Relapse: edema or proteinuria → ++ for 3 consq days
- Frequent: > 3attscks / 6months Or > 4 / yr
- Infrequent.

N.B. : all steroid dependent are relapse But not the reverse

- Treatment of infrequent : as first attack
- 60 mg/m2 is given untill clearance of proteinuria (for 3 days). Not for 4 wks.
- Treatment of frequent : as steroid dependant .
- Steroid dependant :
- Maybe when change to 40 on tapering Or within 24 days after end of ttt .
- **Treatment**
- Start from 60 then 40 then 25 .
- But instead of 10 “maintain them on higher dose and for longer time “.

- eg. 15 for 18 months (thus should be < 1 mg/kg)
- 60mg till clearance then→if> 1 mg/kg :cytotoxic drug

- So cytotoxic drugs :
  - Frequent relapse.
  - Stroid dependant
  - Steroid resistant.
- Biopsies to atypical NS .

Enuresis :

Urinary incontinence in child who is considered mature enough to acquire urinary continence.

Types : Primary VS secondary

- Nocturnal VS diurnal VS both.

N.B. normally day time dryness >> 4 yrs.

Nocturnal dryness >> 6 yrs "boys" .. 5 yrs "girls"

- **Epidemiology :**

- Nocturnal > Diurnal
- Nocturnal (61%) M>F, Diurnal F>M.

- **Causes :**

1. Genetic "+ve family history", #1 cause of 1ry Nocturnal.
2. Psychological eg ; new home, conflicts
   #1 cause of 2ry Nocturnal.
3. Sleep disorders : (adenoid, deep sleeping), also in 2ry Nocturnal.
4. Organic : UTI, DM, DI, constipation
   #1 in 2ry diurnal and Nocturnal.
5. Lack of Nocturnal ADH peak.
6. Neurodevelopmental : eg; neuropathic bladder in spina bifida. #1 cause in 1ry Diurnal and Nocturnal.

- **Investigations :** Urine dipstick :glucose, WBCs,... Etc

- **Treatment :**

1. Explanation and reassurance (avoid punishment)
2. Star chart (positive reinforcement)
3. Pharmacotherapy : - Tricyclic antidepressants "TCA", impairamine. Desmopressin.

Note : 90% recurrence if TX stopped.

4. hypnotherapy
   - **N. B**. all those TX after treating the underlying cause.
     15% will recover each year.

7# Genetics :

Some Definitions:

- hereditary : derived from parents.
- congenital : presents from birth
  >> Not all hereditary are congenital (eg Huntingtons)
  >> Not all congenital are hereditary (eg TORCH.
- familial : run in families ; may be genetic or environmental.
- alleles : different versions of genes.
- Mosiascism : 2 cell lines from one zygote.
- genetically determined diseases :

- Polygenic (multifactorial, complex)
- Mendelian disorders (single gene)
- Chromosomal abnormalities (numeral /structural)
- Usual genetic mechanisms (trinucleotide repeat, unipaternal disomy)

Polygenic :

- Interaction between genetic and environmental factors.

Eg – congenital malformation #1 cause, cleft lip palate, NTDs, pyloric stenosis.

_ common childhood /adult disease. Eg asthma, epilepsy, atherosclerosis.

_ quantitative traits. Eg height , intelligence.

- Tend to run in families, but don't follow normal Mendelian inheretance.
- Distribution in population : normal (Gaussian) Eg pyloric stenosis thickness.
- You get the disease when you reach threshold.
- Recurrence : rate in family is low but increase by :
  -severity of disease (bilateral > unilateral cleft lip).
  -close relationship to the affected person.
  - high number of family members affected.
  -sex difference in prevalence. (N.B Pyloric stenosis = males)

Mendelian inheritance:

AD :
No skip generation (vertical)
No carrier
No male / female difference

- The child of affected parent has 50% chance if getting the disease.
  Eg

- A chondroplasia
- Marfan's
- NF
- Polyposis coli
- Noonan's syndrome
- Huntington disease

Structural (anatomical)

- Myotonic dystrophy
- Ehler danlo's sign
- Tuberous sclerosis
- Osteogenesis imperfect
    - Variation in expression : some are mild disease, others severe.
    - Non pentrance : No c/o or S & S (eg otosclerosis)
    - N. B No family hx
      >> new mutation (eg 80% of achondroplasia)
      >> paternal Mosaicism " normal cells + abnormal cells " (eg Osteogenesis imperfecta)

AR	X-linked Recessive
• **Skip generation (horizontal).** • **+ve carrier.** • **No male/female difference.** • **Children of two carriers have 25% chance getting the disease. 25% normal, 50% carriers.** • **Consanguinity increase the risk of rare trait here eg : cystic fibrosis, CAH, sickle cell, errors of metabolism "eg PKU"** • **Enzymatic.**	• Skip generations. • Females are carriers, males affected. • Female carrier : 50% of her sons are diseased 50% of her daughters are carriers. • Male affected : No son affection (0%) 100% of his daughters are carriers • Eg Color blindness Hemophilia Duchene/ Becker Fragile X syndrome G6PD deficiency Fragile X syndrome. **N.B.** diseased females => lyonization tumor.

X-linked dominant:

- Eg :hyperphosphatemic (vit D resistant) rickets.
  Incontientia pigmenti
  Alport's (multiple modes of inheretance) Rott's .

Chromosomal abnormalities :

- In number (aneuploidy)

#Down syndrome :

- **Causes:**
  - Non disjunction 95%
  - Roberstonian translocation*
  - Mosiascism



*In acrocentric chromosomes (13,14,15,21,22).

Most common in down 14,21

- **C/O:**

- Hypotonia
- Characteristic face : epicanthic folds, upslanting of palpepral fissures, flat nasal bridge, small mouth, large tongue (protruding), low set ears, flat occipt.
- Hand: single Palmar crease, short broad hands
- Feet: sandal gab
- At birth: polycythemia (may need treatment)
- Leukomoid reaction (self limiting)
- Cardiac: Cushing defect, AV canal defect.
- GI : D-atresia, Imperforated anus, hischsprung disease, annular pancreas.
- **later :**
  Mental retardation
  Increased risk of infection
  Increased risk of leukemia, alzehimer's, hypothyroidism, atalantoaxial instability.
  - **screening:** Women:
    - >35 years :amniocentesis or chorionic villous sampling
    - <35 years : nuchal translucency scan →Serum alph afetoprotein(low) if +ve -> aminocentesis
  - **Recurrence risk**
    - 1% added to age specific risk < 35 years
    - Just age specific risk > 35 years
    - If child has Rob-translocation : do Chromosomal analysis for parents.
    - If +ve balanced translocation in one parent : risk is higher if mother was the one with the B-translocation.

Edward's syndrome (tri 18):

>90% aborted

10% reach 1st year.



- **C/O**
  - Rocker bottom feet
  - Prominent occipt (VS down)
  - Overriding of fingers, clenched fist

Patau syndrome (tri 13):

- **C/O**
  - Midline defects, cut is aplesia (punched out skull), cyclopia (one orbit), cebocephaly (single nostril, hole proncephaly
    - In both tri 18 and tri 13 >> congenital heart disease.
    - N. B Recurrence risk is low except for balanced translocation.



Klienfenter syndrome : (47, XXY)

- Normal before puberty
- After puberty : small gonads, no 2ry sex characteristics, tall stature, gynecomastia, infertility "first presentation"
- Treatment : testesterone to develop 2ry sex characteristics.

Edward's and klineflter's are genetic cause of infertility in males.

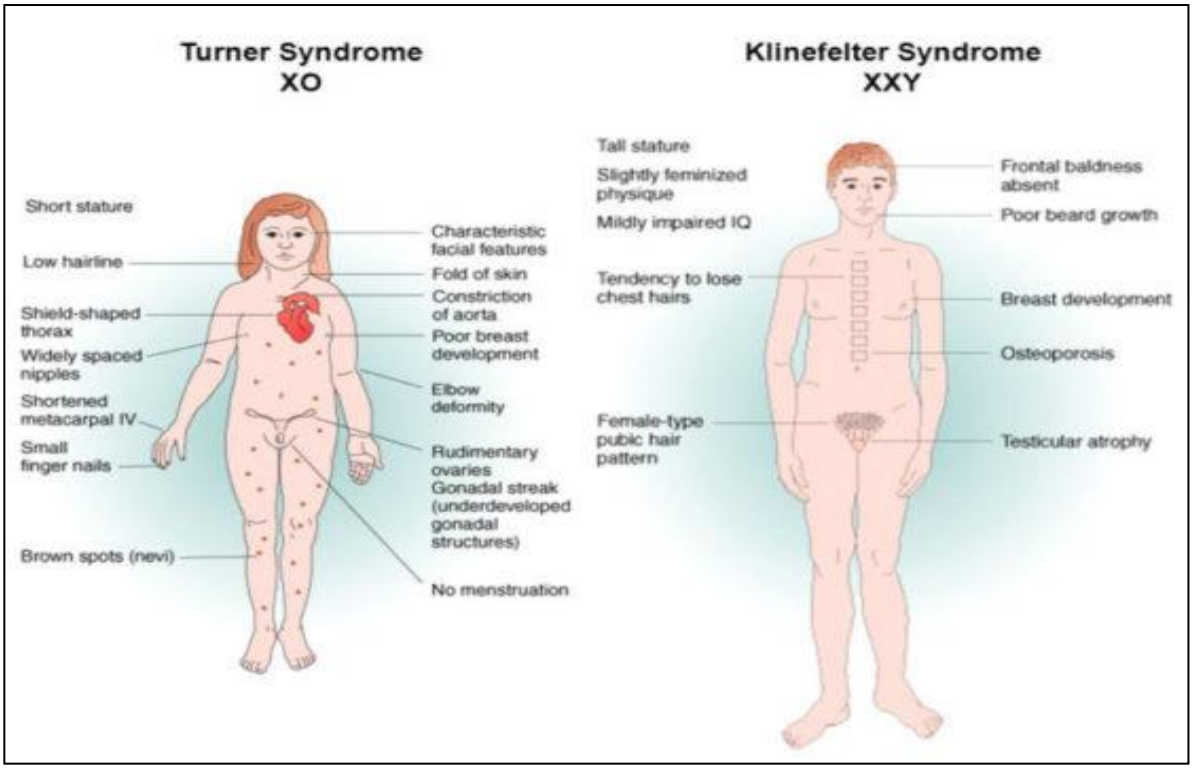
Turner's syndrome (45 XO) :

- 99% aborted.
- #1 aneuploidy in conceptus
- Normal mentation
- At birth : lymphedema

- **C/O** :

- Short stature " main feature"
- Webbed neck, hygroma
- Sheild like chest (wide nipples)
- Wide carrying angle (cabitus valgus)
- Increased risk of hypothyroidism
- Heart : coartication of aorta, Bicuspid aortic valve

- **Treatment :** estrogen for 2rd sex characteristics, no role in infertility. + GH
  - Incidence not increased with increased maternal age.
  - Hypothyroidism : down's / Turner's.



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Ola Salih

Rayan Othman

Zuhal Humaidah

إن ما يهون علينا دائماً أننا سنتذكر لاحقاً هذه اللحظة الثقيلة

التي تمر من فوقنا ونتنهد تنهيدة المنتصر